VARUN K HARI

Python Backend Developer | FastAPI | AI Applications
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PROFESSIONAL SUMMARY

Python Backend Developer with 1 Year startup experience building AI-powered web applications and backend systems using FastAPI, Django, and PostgreSQL. Worked on OCR-based document processing, REST API development, and LLM-integrated applications including RAG systems and automation workflows. Familiar with deploying backend services using Docker and AWS. Passionate about building scalable backend applications and AI-assisted software solutions.

TECHNICAL SKILLS

- **Backend & Web :** Python, FastAPI, Django, Rest APIs, React, PostgreSQL, MySQL
- **AI/ML**: RAG Systems, LangChain, Ollama, Prompt Engineering, OCR workflows
  Whisper, Streamlit
- **Tools &Databases:** Docker, Git/GitHub, AWS (EC2, S3), Postman

PROFESSIONAL EXPERIENCE

Jr. Machine Learning Engineer (Backend & AI Systems) – Codedu Software Technologies | 2025 - 2026

- Built backend APIs using FastAPI and Django for hospital management and patient workflow systems
- Developed OCR-based patient onboarding workflows using MRZ extraction and regex-based validation
- Worked on document processing and structured data extraction pipelines for healthcare applications
- Integrated REST APIs for billing, insurance validation, and patient management modules
- Assisted in developing backend modules for a microservices-based hospital management platform
- Containerized backend applications using Docker and deployed services on AWS EC2

Banking Operations Associate – Federal Bank | Jul 2024 – Apr 2025

- Assisted in daily banking operations and customer service activities
- Maintained operational documentation and supported branch workflow processes
- Worked in a fast-paced environment requiring communication and problem-solving skills

Safety Compliance Officer(HSE)| Jul 2023 – Jun 2024

- Conductedworkplacesafetyauditsand inspections.
- Maintainedsafetycompliancedocumentation and regulatory reporting.
- Ensuredadherencetoworkplacesafety standards and operational risk management.

TECHNICAL PROJECTS

Patient Care Hospital Management System (PCHMS)
- Built a full-stack hospital automation platform with microservice architecture
- Implemented OCR-based patient registration using ID card scanning
- Designed intelligent extraction system using MRZ + layout-based parsing
- Automated workflows: appointment, billing, insurance validation, room allocation
- Integrated role-based system for doctors, nurses, admin, and patients

AI Voice Assistant & RBI Policy Retrieval System

• Developed an AI-powered banking voice assistant using Langflow integrated with Llama 3.2 LLM.
• Implemented speech-to-text using Whisper and text-to-speech using Edge-TTS to enable real-time voice-based interaction.
• Built a domain-specific Retrieval-Augmented Generation (RAG) pipeline using Langflow and ChromaDB for RBI policy retrieval.
• Designed the system to run locally without internet dependency, improving data privacy and performance.

Intelligent E-Commerce Platform

• Built a full-stack e-commerce web application using Django, React, and PostgreSQL.
• Implemented user authentication, product catalog, order management, and payment-ready architecture.
• Integrated a machine learning recommendation system to suggest products based on user browsing behavior.
• Developed REST APIs for frontend-backend communication.

Course Selling Platform (Learning Management System)

• Developed a complete Python Full Stack LMS platform using Django, HTML, CSS, JavaScript, and PostgreSQL.
• Implemented features such as course enrollment, instructor course uploads, student dashboards, and secure authentication.
• Built backend APIs and database models to manage courses, users, and transactions.
• Created a student analytics dashboard using Pandas and Matplotlib to visualize learning performance and engagement.

EDUCATION

- **Advanced Certification in Python Full Stack, Data Science ML & Integrated AI** — Codedu Software Technologies (Apr 2024 – Feb 2025)
- **Bachelor of Science (HSE)** — Annamalai University (Jun 2020 – May 2023)

CERTIFICATIONS

- Generative AI with Large Language Models – DeepLearning.AI
- Introduction to Artificial Intelligence – IBM (Coursera)